

# Corporate Finance

## Lecture 5: Investor Behavior and Capital Market Efficiency

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# Introduction

- ▶ Is it possible to systematically “beat” the market?
- ▶ Bill Miller, as a fund manager of Legg Mason Value Trust:
  - ▶ Outperformed the market each year between 1991 and 2005.
  - ▶ In 2007 and 2008, the fund lost almost 65% of its value.
- ▶ According to the CAPM, it should be impossible to achieve a performance (= Sharpe ratio) superior to the one of the market in a persistent manner!

# Outline

- A. Competition and capital markets (Chap 13.1)
- B. Information and rational expectations (Chap 13.2)
- C. The behavior of individual investors (Chap 13.3)
- D. Systematic trading biases (Chap 13.4)
- E. The efficiency of the market portfolio (Chap 13.5)
- F. Style-based techniques and the market efficiency debate (Chap. 13.6)
- G. Methods used in practice (Chap. 13.8)

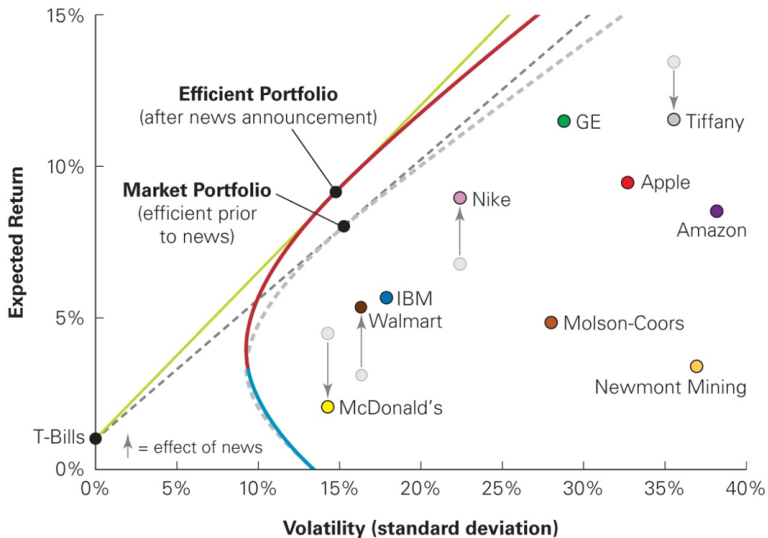
## A. Competition and capital markets

- ▶ Necessary condition to beat the market?
  - ▶ The expected returns on some investments need to exceed their required returns.
- ▶ Difference between expected return and required return: alpha of an asset

$$\alpha_i = E(R_i) - r_i = E(R_i) - (r_f + \beta_i (E(R_M) - r_f)).$$

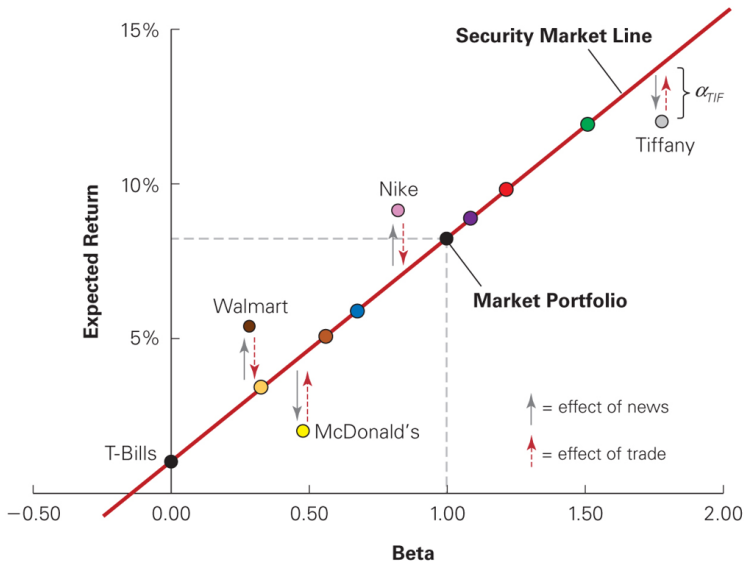
- ▶ If the market portfolio is efficient, all the securities and all the portfolios are on the Security Market Line.  $\Rightarrow$  Their alphas are zero.

# An inefficient market portfolio: example



Market becomes inefficient after news announcement which lowers

# Deviations from Security Market Line: example



You can improve upon market portfolio by buying (selling) stocks

## B. Information and rational expectations

- ▶ Hypothesis of the CAPM  $\Rightarrow$  all investors have access to the same information set
- ▶ Unrealistic in practice!



VS.



- ▶ But “naïve” investors do not necessarily lose out:

## Example: how to avoid to be beaten by informed investors?

- ▶ Jérôme does not have access to any information regarding stocks. Nevertheless, he knows that the other investors have a great deal of information and use that information when they construct their portfolio.
- ▶ This information asymmetry is a concern to Jérôme, because he thinks it will cause his portfolio to underperform the portfolio of the other investors.  
  
⇒ How can he guarantee that his portfolio will do as well as that of the average investors?

# Rational expectations and the inefficiency of the market portfolio

The market portfolio can be inefficient (so it is possible to beat the market) only if a significant number of investors either

- ▶ Do not have **rational expectations** so that they misinterpret information and believe they are earning a positive alpha when they are actually earning a negative alpha,

OR

- ▶ Care about aspects of their portfolios other than expected returns and volatility, and so are willing to hold inefficient portfolios of securities.

## C. The behavior of individual investors

### ► Underdiversification

- In 2001, for households that held stocks, median number of stocks held was 4, and 9 out of 10 held less than 10 stocks.  
Possible explanations:

- Familiarity bias
- “Keeping up with the Joneses” (relative wealth concerns)

### ► Overconfidence

- Leads to excessive trading  $\Rightarrow$  Reduces the performance due to transaction costs.

- Underdiversification and excessive trading violate key predictions of CAPM, but does this imply that remaining conclusions of CAPM are also invalid?
- Not necessarily, if departures from CAPM are random.  $\Rightarrow$  To have an impact, biases need to lead to systematic deviations from rational behavior.

## D. Systematic trading biases

- ▶ Disposition effect
  - ▶ Tendency to sell winners too early and hang on to losers too long.
- ▶ Investor attention, mood, and experience
- ▶ Herd behavior (imitate each other's actions)
  - ▶ Informational cascade effect: ignore own information and hope to profit from information of others.
  - ▶ Relative wealth and performance concerns.

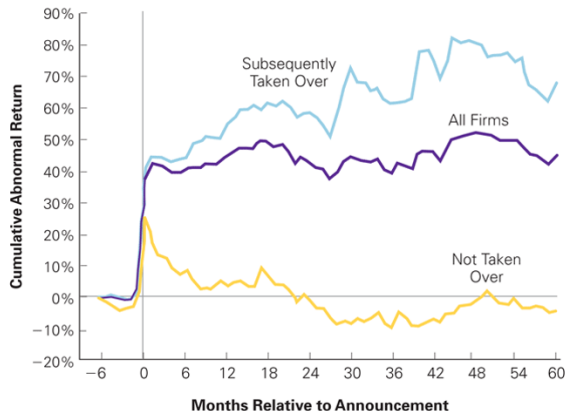
⇒ Consequences: Biases will affect prices if:

- ▶ Biases are sufficiently pervasive and persistent.
- ▶ Competition to exploit these arbitrage opportunities is limited.

## E. The efficiency of the market portfolio

Example for trading on news: Takeover offers

- If you could predict whether firm would ultimately be acquired or not, you could earn trading profits.



Source: Adapted from M. Bradley, A. Desai, and E. H. Kim, "The Rationale Behind Interfirm Tender Offers: Information or Synergy?" *Journal of Financial Economics* 11 (1983) 183-206.

Returns to holding target stocks subsequent to takeover announcements

# Performance of fund managers

## ▶ Fund Manager Value-Added

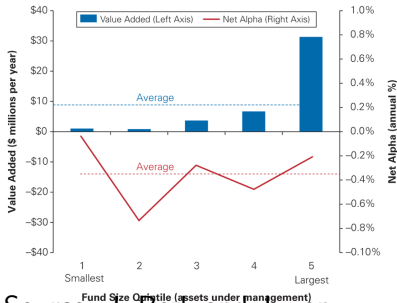
- ▶ Before fees and adjusting for assets under management, the average mutual fund adds value.
- ▶ However, the median mutual fund destroys value.
- ▶ Most fund managers appear to trade so much that their trading costs exceed the profits from any trading opportunities they may find.

## ▶ Return to Investors

- ▶ Numerous studies report that the actual alpha (net of fees) to investors of the average mutual fund is negative.
- ▶ Superior past performance is not a good predictor of a fund's future ability to outperform the market.

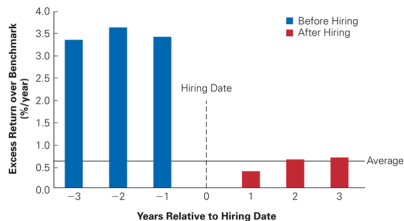
# Performance of fund managers (con't)

## Manager Value Added and Investor Returns for U.S. Mutual Funds



Source: J. Berk and J. van Binsbergen, "Measuring Managerial Skill in the Mutual Fund Industry," *Journal of Financial Economics* 118 (2015): 1-20.

## Before and After Hiring Returns of Investment Managers



Sources: A. Goyal and S. Wahal, "The Selection and Termination of Investment Management Firms by Plan Sponsors," *Journal of Finance* 63 (2008): 1805-1847 and with J. Busse, "Performance and Persistence in Institutional Investment Management "

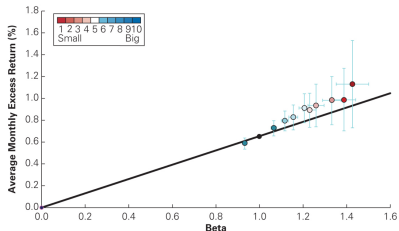
# The winners and the losers

- ▶ The average investor earns an alpha of zero, before including trading costs.
- ▶ Beating the market should require special skills or lower trading costs.
  - ▶ Because individual investors are likely to be at a disadvantage on both counts, the CAPM wisdom that investors should “hold the market” is probably the best advice for most people.

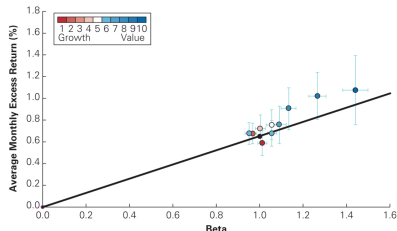
## F. Style-based techniques and the market efficiency debate

- ▶ Excess return and market capitalizations (Size effect)
  - ▶ Small market capitalization stocks have historically earned higher average returns than the market portfolio, even after accounting for their higher betas.
- ▶ Excess return and book-to-market ratio (B/M effect)
  - ▶ High book-to-market (value) stocks have historically earned higher average returns than low book-to-market (growth) stocks.
- ▶ Momentum
  - ▶ Strategy: buy stocks that have had past high returns and (short) sell stocks that have had past low returns.
- ▶ Empirical evidence:
  - ▶ Possibility of a data snooping bias: given enough characteristics, it will always be possible to find some characteristic that by pure chance happens to be correlated with the estimation error of average returns.

# Empirical evidence (con't)



Excess Return of Size  
Portfolios, 1926-2015



Excess Return of  
Book-to-Market Portfolios,  
1926-2015

Data is from Ken French's datalibrary: [http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data\\_library.html](http://mba.tuck.dartmouth.edu/pages/faculty/ken.french/data_library.html).

# Implications of positive-alpha trading strategies

- ▶ The only way positive-alpha strategies can persist in a market is if some barrier to entry restricts competition.
  - ▶ However, the existence of these trading strategies has been widely known for more than 20 yrs.
- ▶ Another possibility is that the market portfolio is not efficient, and therefore a stock's beta with the market is not an adequate measure of its systematic risk.
- ▶ Proxy error: true market portfolio may be efficient, but our proxy may be inaccurate.
- ▶ Behavioral biases: may induce investors to hold inefficient portfolios.
- ▶ Alternative risk preferences and non-tradable wealth: may induce investors to choose inefficient portfolios (in the mean-variance sense).

## G. Methods used in practice

- ▶ There is no clear answer to the question of which technique is used in practice—it very much depends on the organization and the sector.
- ▶ In addition, all the techniques covered are imprecise.
- ▶ A survey of CFOs found that
  - ▶ 73.5% of the firms used the CAPM to calculate the cost of capital.
  - ▶ 40% used historical average returns.
  - ▶ 16% used the dividend discount model.
  - ▶ Larger firms were more likely to use the CAPM than were smaller firms.

# Quiz

- ▶ If investors buy a stock with a positive alpha, what is the likely effect on its price and expected return?
- ▶ How can uninformed investors guarantee themselves a non-negative alpha?
- ▶ Why is the high trading volume observed in markets inconsistent with the CAPM equilibrium?
- ▶ What are some of the systematic trading biases to which individual investors fall prey?